DATA Merging

Assignment 1:
Question: Merge the two datasets into a single dataset patient_info .
What happens if you don't use the BY statement?

```
data demo;
input ID $ Name $;
datalines;
101 John
102 Mary
;
run;

data vitals;
input ID $ Height Weight;
datalines;
101 170 65
102 160 55
;
run;
```

Assignment 2:
Sort and merge these two datasets by ID using BY ID.
Print the final output and verify both Age and Gender are matched correctly.

```
data a;
input ID $ Age;
datalines;
101 25
102 30
;
run;

data b;
input ID $ Gender $;
```

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```
datalines;
101 M
102 F
;
run;
```

Assignment 3:
Perform a many-to-one merge to assign the hospital name to each patient visit. What happens for ID 103?

```
data patients;
input ID $ VisitDate :date9.;
format VisitDate date9.;
datalines;
101 01JAN2024
102 05JAN2024
103 10JAN2024
;
run;

data hospital;
input ID $ Hospital $;
datalines;
101 Apollo
102 Fortis
;
run;
```

Assignment 4:
Use the SET statement to concatenate the two site datasets into a new dataset all_sites.

```
data site1;
input SubjID $ Value;
datalines;
A01 10
A02 20
;
```

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```
run;

data site2;
input SubjID $ Value;
datalines;
B01 15
B02 25
;
run;
```

Assignment 5:
Sort both datasets by SubjID and interleave them using the SET statement with BY SubjID.
What is the difference from a simple concatenation?

Assignment 6:
Merge both datasets and create a variable source_flag which should show:
'Both' if the ID exists in both
'Only_DM' if only in dm
'Only_EX' if only in ex

```
data dm;
input ID $ Age;
datalines;
101 34
103 30
;
run;

data ex;
input ID $ Dose;
datalines;
101 50
102 75
;
run;
```

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Assignment 7:
Using IN= variables, perform a full outer merge to retain all observations from both datasets.
Highlight missing values.

Assignment 8:
Use the UPDATE statement to apply the update to the original dataset.
What is the result for each ID?

```
data original;
input ID $ Status $;
datalines;
101 Enrolled
102 Withdrawn
;
run;

data updates;
input ID $ Status $;
datalines;
102 Completed
;
run;
```

Assignment 9:
Merge the datasets using ProductID to create a new dataset order_details that includes the ProductName.

```
data orders;
input OrderID ProductID Quantity;
datalines;
1001 501 2
1002 502 1
;
run;

data products;
input ProductID ProductName $;
datalines;
```

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```
501 Aspirin
502 Tylenol
;
run;
```

Assignment 10:
Merge lab and demo data (keep all lab records).
Handle missing values for AGE.

```
data lab;
input USUBJID $ LBTEST $ RESULT;
datalines;
001 Glucose 90
002 BP 120
003 BP 130
;
run;

data demo;
input USUBJID $ AGE;
datalines;
001 45
003 50
;
run;
```

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